

Javascript

Class & objects, inheritance, overriding, String

```
<!DOCTYPE html>

<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Strings</title>
</head>
<body>
  <h1>String</h1>

  <script>

    // mutable -> changeable
    // immutable -> unchangeable
    // string -> immutable

    let name = "Hello welcome";
    let college = 'VIT';

    let course = `LLB`; // template literal -> grave accent

    let newKeyword = new String("Naren");

    let details = {name, college, course};

    console.log(details);

    console.log(name, college, course);

    document.writeln(name);

    // string methods

    // document.writeln(name.length);
    console.log(name.length);

    console.log(name.indexOf('a'));

    console.log(name.lastIndexOf('a'));
```

```

console.log(name.substring(3,0));

// 3 => n-1 => 3-1 => 2

console.log(name.slice(3,0));

console.log(name.slice(-1));

console.log(name.substring(-4));

// charAt
console.log(name.charAt(2));

// replace
console.log(name.replace("welcome", "FIIT"));

// startswith

console.log(name.startsWith("come"));

// endswith

console.log(name.endsWith("come"));

// split

console.log(name.split(" "));

// lowercase
console.log(name.toLowerCase());
// uppercase
console.log(name.toUpperCase());
// trim

console.log("  hello welcome  ".trim());

// includes

console.log(name.includes("e"));
// class & objects -
class Bank{

    balance =10;

    constructor(name, age, college){
        console.log(`
        Constructor is called->
        ${name}

```

```

        ${age}
        ${college}`);
    }

    display(){
        console.log("Bank class");
    }
}
//inheritance
class Account extends Bank{
    constructor(){
        super();
        super.display();
        console.log("accounts class constructor is called");

    }
    //overriding
    display(){
        console.log("Account class");
    }
}
let account = new Account();

console.log(account.balance);
account.display();

let bank = new Bank("Maha", 20, "kasc");

console.log(bank.balance);
bank.display();

</script>

</body>
</html>

```

Theory Notes

1. Strings in JavaScript

- **Immutable** → once created, cannot be changed directly.
- Can be defined using:
 - Double quotes "Hello"
 - Single quotes 'Hello'
 - Template literals (backticks) `Hello` → allow interpolation \${variable}.
- **Common methods:**
 - `length` → returns string length.
 - `indexOf()`, `lastIndexOf()` → find position of substring.

- `substring(start, end)` → extracts part of string.
- `slice(start, end)` → extracts part, supports negative indices.
- `charAt(index)` → returns character at position.
- `replace(old, new)` → replaces substring.
- `startsWith(), endsWith()` → check beginning/end.
- `split(separator)` → splits into array.
- `toLowerCase(), toUpperCase()` → case conversion.
- `trim()` → removes whitespace.
- `includes()` → checks if substring exists.

2. Classes and Objects

- **Class** → blueprint for creating objects.
- **Constructor** → special method called when object is created.
- **Inheritance** → one class extends another (extends).
- **Method overriding** → child class redefines parent method.
- **super()** → calls parent constructor or methods.

Coding Round Questions

Strings

1. Create a string "Hello World" and print its length.
2. Write a program to replace "World" with "JavaScript".
3. Extract the substring "Hello" using `substring()` and `slice()`.
4. Convert a string to uppercase and lowercase.
5. Split a sentence into words and print them as an array.
6. Check if a string starts with "Hello" and ends with "Script".
7. Trim extra spaces from " Welcome " and print result.

Classes & Inheritance

8. Create a class `Bank` with a property `balance` and a method `display()`.
9. Extend `Bank` into a class `Account` that overrides `display()`.
10. In `Account`, call the parent method using `super.display()`.
11. Create an object of `Account` and demonstrate inheritance.
12. Create another object of `Bank` with constructor parameters and print them.